

Guesstimate 2

Estimating the Number of EV Charging Stations Required in Bengaluru

Business Scenario

VoltCharge Solutions is a startup specializing in electric vehicle charging infrastructure. The company plans to expand into Bengaluru, where EV adoption has grown rapidly over the last few years. Before making investments in charging infrastructure, the founders want to estimate the potential demand for charging stations.

The company has approached your consulting team to estimate how many public EV charging stations Bengaluru may require over the next three years. Since detailed market data is unavailable, the estimate should be based on logical assumptions regarding the city's population, EV ownership, charging frequency, and station capacity.

Your analysis will help the company determine whether Bengaluru is an attractive market and support decisions related to investment, expansion, and infrastructure planning.

Question

Estimate the number of **public EV charging stations required in Bengaluru** over the next three years.

Step 1: Understand the Problem

Objective

Estimate the number of **public EV charging stations** required in Bengaluru over the next **3 years**.

Scope

Assume we are estimating **only public charging stations**.

Exclude:

- Home charging
 - Office charging
 - Private fleet charging
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Step 2: Build the Approach

Instead of guessing directly, divide the problem.

Number of Public EV Charging Stations = Total EVs in Bengaluru ÷ Number of EVs served by one charging station

Step 3: State Assumptions

Parameter	Assumption	Reason
Bengaluru Population	1.5 Crore	Approximate city population
Vehicle Ownership	1 vehicle for every 3 people	Urban city with high vehicle ownership
Total Vehicles	50 Lakh	Calculated
EV Penetration	10%	Growing EV adoption
Public Charging Dependency	40%	Many users charge at home/work
EVs served per Charging Station	100 EVs	Average daily utilization

Step 4: Calculate Total Vehicles

Population = 1.5 Crore

Vehicle ownership = 1 vehicle per 3 people

Total Vehicles = 1.5 Crore ÷ 3

= **50 Lakh Vehicles**

Step 5: Estimate Total EVs

Assume

10%

of vehicles are electric.

50 Lakh × 10%

=

5 Lakh EVs

Step 6: Estimate Public Charging Demand

Not every EV depends on public charging.

Many owners charge:

- At home
- At office

Assume only

40%

regularly require public charging.

$5 \text{ Lakh} \times 40\%$

=

2 Lakh EVs

Step 7: Estimate Charging Capacity

Assume

One charging station can comfortably serve

100 EVs.

Therefore,

Required Stations

=

$2,00,000 \div 100$

=

2,000 Charging Stations

Step 8: Validate the Estimate

Ask yourself,

Does this number seem realistic?

Considering

- Airports
- Malls
- Offices
- Highways
- Residential Areas

Approximately **2,000 public charging stations** appears to be a reasonable estimate for a city the size of Bengaluru.

Step 9: Factors Affecting the Estimate

The estimate may vary based on:

- Faster EV adoption
 - Government subsidies
 - Charging speed
 - Battery technology
 - Home charging availability
 - Commercial fleet demand
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Step 10: Final Recommendation

Based on the assumptions made, Bengaluru may require approximately **2,000 public EV charging stations** over the next three years to support the growing electric vehicle ecosystem.

Before expansion, VoltCharge Solutions should conduct location-wise demand analysis to identify high-traffic areas such as business districts, shopping centers, metro stations, highways, and residential hubs to maximize utilization.